FaceTrack

Face Recognition Attendance System

Complete User Guide & Platform Manual

Step-by-step instructions for all user roles

Version 1.0 · 2026

https://attendace-system-david.vercel.app/

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1 Platform Overview

FaceTrack is a web-based attendance management system that uses **face recognition** to identify students during lectures. Instead of calling a roll, a lecturer simply points a webcam at the class and FaceTrack marks each face it recognises within seconds — with no queuing, no sign-in sheet, and no proxy attendance.

> 🌐 **Live Platform**
>
> Access the system at: https://attendace-system-david.vercel.app/

How It Works

🎓 Enrol

An administrator registers each student once, capturing 3 face samples with a webcam. The system stores 128-point face vectors (not photos) for privacy.

📸 Capture

During a lecture, the lecturer opens the live capture screen. The camera scans the room every ~0.9 seconds, recognising and marking faces automatically.

📊 Report

After sessions, the system computes each student's attendance percentage and flags who is below the required threshold for exam eligibility.

User Roles

ROLE	WHO THEY ARE	WHAT THEY CAN DO
Admin	Department officer / Registrar	Full access: manage students, courses, sessions, staff accounts, reports
Lecturer	Course lecturer	Start sessions, capture attendance on assigned courses, view reports. Cannot create students or manage staff.

> ⚠️ **Important:**
>
> Students do *not* have login accounts. They are identified purely by face recognition during an active session. Students can only be added by an Admin.

Navigation Overview

▤ Dashboard

Stats overview, open sessions, recent records

👥 Students

Student list, registration, face enrolment, profiles

📚 Courses

Course management and class lists

🎥 Lecture Sessions

Start, run, and close lecture attendance sessions

📊 Attendance Report

Per-course registers, percentages, CSV export

🔑 Staff Accounts Admin only

Create and manage lecturer/admin accounts

2 Getting Started — Login & Credentials

Accessing the Platform

Open a web browser (Chrome, Edge, or Safari recommended) and navigate to:

🔗 **Live URL**

https://attendace-system-david.vercel.app/

You will be directed to the **Login page** if you are not already signed in.

Login Credentials

The system ships with two default staff accounts. Use these to sign in:

🔐 **DEFAULT LOGIN ACCOUNTS**

ROLE	EMAIL ADDRESS	PASSWORD	ACCESS LEVEL
Administrator	admin@example.edu	admin1234	Full system access
Lecturer	lecturer@example.edu	lecturer1234	Courses & sessions only

⚠️ **Security Notice:**

These are demo credentials. If you create real staff accounts (Section 8), ensure you use a strong password (minimum 6 characters). The system does not enforce password complexity, so use good judgment.

How to Log In

1. **Open the login page**
   Visit `https://attendace-system-david.vercel.app/`. You will see a two-panel screen: a dark blue branding panel on the left and a login form on the right.
2. **Enter your email address**
   Type your staff email in the "Email address" field. For the admin account: `admin@example.edu`
3. **Enter your password**
   Type your password. For the admin account: `admin1234`
4. **Click "Sign in"**
   You will be redirected to the **Dashboard** automatically.

Session Duration

Your login session lasts **8 hours** from the time you sign in. After that, you will be automatically signed out and redirected to the login page. Simply sign in again to continue.

Signing Out

At the bottom of the sidebar (visible on all pages), you will see your name, role, and email. Click the **"Sign out"** button to log out securely.

> ✅ **After Login:**
>
> The sidebar on the left shows only the sections available to your role. Admins see all 6 items including "Staff accounts". Lecturers see 5 items (no Staff accounts link).

3 Dashboard

The Dashboard is the first screen you see after logging in. It gives you a quick overview of what is happening across the system right now.

Dashboard Cards (Top Row)

CARD	WHAT IT SHOWS	INDICATOR
Registered students	Total number of students in the system. Shows how many are missing face templates.	Amber if any students lack faces; green if all enrolled
Courses	Number of courses visible to you (all for admin, assigned only for lecturers)	Blue
Open sessions	How many lecture sessions are currently live (attendance can be taken)	Green if sessions are running; grey if none
Marked today	Total attendance records created today. Split into face-recognised vs manually marked.	Grey

Open Lecture Sessions Panel

Shows any sessions currently in progress. Each row displays the course code, session title, start time, and venue. Click **"Open camera"** to go directly to the live capture screen for that session.

> 💡 **Quick access:**
> If no sessions are running, a "Start a session" button takes you straight to the Sessions page to begin one.

Recent Attendance Panel

Displays the 8 most recent attendance records across all courses. Each entry shows the student's name, course code, time marked, and whether they were marked `present` or `late`.

Typical Use: Starting Your Day

📋 Morning Workflow

1. Log in and land on the Dashboard
2. Check the "Open sessions" tile — if it shows a number, someone has already started a session
3. If you need to take attendance: click "Start a session" or navigate to Lecture Sessions
4. If you need to check on a student: click their name in Recent Attendance, or go to Students

4 Students Module

4.1 — Viewing the Student List

Click **Students** in the sidebar. You will see a table of all registered students, showing:

- Student name (with a small avatar thumbnail if face has been enrolled)
- Matriculation number
- Department
- Level (100–500)
- Face template status — **N samples** (enrolled) or **Not enrolled**

Use the **Search box** at the top to filter by name, matric number, or department.

> ⚠ **Important:**
>
> Students marked "Not enrolled" cannot be recognised by the camera. They must have their face enrolled (Section 4.2) before attendance can be taken automatically.

4.2 — Registering a New Student & Face Enrolment

Admin only — The "+ Register student" button is only visible to admins.

1. **Click "+ Register student"**
   (top-right of Students page)

2. **Fill in the Student Details form (left panel)**

   Required: `Matriculation number` (e.g. `CSC/2021/034`) and `Full name`.
   Optional: Email, Department, Level.

3. **Open the Face Enrolment panel (right panel)**

   Click **"Turn on camera"**. Your browser will ask for camera permission — allow it.

4. **Capture 3 face samples**

   The panel will prompt 3 poses one at a time:
   ① *"Look straight at the camera"*
   ② *"Turn your head slightly to the left"*
   ③ *"Turn your head slightly to the right"*
   Click **"Capture sample"** for each pose. A green box will flash on the detected face. Each captured thumbnail appears below.

5. **Click "Save student"**

   The student is saved. You are returned to the Students list.

💡 **Camera tips for best results:**

- Ensure the room is well-lit. Natural or overhead light works best.
- The student should be 50–100 cm from the camera.
- Remove hats, sunglasses, or anything covering the face.
- The counter shows `0/3`, `1/3`, `2/3`, `3/3` as you progress.
- If capture fails, check the error message (e.g. "No face detected").
- Use "Clear samples" to restart if the captures look poor.

4.3 — Viewing a Student Profile

Click any student's name to open their profile. The profile shows:

- **Personal details** — matric number, name, email, department, level
- **Enrolled courses** — a list of courses the student is registered on
- **Attendance history** — the last 50 attendance records: date, course, status, method (face/manual), match distance

4.4 — Re-enrolling a Student's Face

If a student's face samples are poor (bad lighting, significant appearance change), an admin can re-enrol them from the student profile page.

1. Open the student's profile page
2. Click **"Re-enrol face"**
3. Capture 3 new samples following the same process as initial enrolment
4. Save — the old templates are replaced

⚠️ **Deleting a Student:**

The delete button on the student profile is permanent. It removes the student record, all attendance records, and removes them from all courses. This cannot be undone.

5 Courses Module

Click **Courses** in the sidebar. Admins see all courses; lecturers see only the courses assigned to them.

Each course card shows:

- Course code (e.g. `CSC 401`) and title
- Number of registered students
- Level and credit units
- Required attendance percentage (default 75%)
- Assigned lecturer

5.1 — Creating a Course Admin only

1. **Click "+ New course"**
   (top-right button). An inline form expands.
2. **Fill in the form:**

FIELD	REQUIRED?	NOTES
Course code	Yes	e.g. `CSC 401` — automatically uppercased, must be unique
Course title	Yes	e.g. `Software Engineering`
Level	No	100, 200, 300, 400, or 500
Credit unit	No	Default: 3
Lecturer	No	Select from existing staff accounts
Department	No	e.g. `Computer Science`
Required attendance (%)	No	Default: 75. Students below this are flagged "Below requirement" in reports.

3. **Click "Create course".**
   The new course card appears in the grid.

5.2 — Managing Students on a Course

Click a course card to open the **Course Detail page**.

Adding students to a course

1. On the course detail page, click
   "Add students"
   (or use the student search/add panel)
2. Search for students by name or matric number and select them
3. Click

"Add selected students"

— they appear in the course's student list

> ⚠️ **Required before starting a session:**
>
> A course must have at least one registered student before a lecture session can be started. Attempting to start a session on an empty course will return an error.

Removing a student from a course

On the course detail page, find the student in the list and click the **remove** (×) button next to their name. This removes them from the course's class list but does not delete their student record or historical attendance data.

Starting a session from the course page

The course detail page also has a **"Start lecture session"** button — a convenient shortcut to launch a session for that specific course. It takes you directly to the Sessions page with the course pre-selected.

Viewing the course report

The **"View report"** button on a course detail page navigates to the Attendance Report for that course.

6 Lecture Sessions

This is the core of the system. A **session** represents a single lecture sitting. Attendance can only be captured while a session is `open`. Once `closed`, the register is locked and cannot be modified except by deleting individual records (admin only).

6.1 — Starting a Session Admin & Lecturer

1. **Click "Lecture sessions"**
   in the sidebar
2. **Fill in the Start Session form at the top of the page:**

FIELD	REQUIRED?	NOTES
Course	Yes	Select from the dropdown. Only courses with ≥1 student are listed.
Venue	No	e.g. `LT1`, `Room 204`
Late after (mins)	No	Default: 15. Students recognised after this many minutes from the start are marked `late` instead of `present`.

3. **Click "Start session & open camera".**
   You are immediately taken to the live capture screen.

> ⚠ **One session at a time:**
> Only one session per course can be open at a time. If you try to start a session for a course that already has an open one, the system will redirect you to the existing open session instead.

6.2 — Live Face-Recognition Capture Screen Admin & Lecturer

This is the most important screen in the system. It consists of two panels: the camera view on the left and the real-time register on the right.

Starting capture

1. **Click "Start capture".**
   The browser may ask for camera permission — allow it.
2. The camera turns on. A `● SCANNING` indicator appears in the top-left corner of the video.
3. Every ~0.9 seconds, the system scans every face visible in the frame.

Understanding the face boxes (overlay colours)

- **Green box** — Student just recognised and marked present (new record)
- **Blue box** — Student recognised but already marked in this session (no duplicate created)
- **Red box** — Face detected but not recognised (not enrolled, not on this course, or lighting issue)

The register panels (right side)

- **Present counter:** Shows how many students are marked vs. total on course (e.g. 8/10, 80%)

- **Marked present list:** Students already marked, with the time, method, and status badge
- **Not yet marked list:** Students still absent — each has a "Mark manually" button as a fallback

"Last frame" panel (below camera)

Shows what the most recent scan detected: for each face found, it shows the student name, matric number, match distance, and whether they were just marked or were already recorded.

> 💡 **Best practices for capture:**
>
> - Ask students to sit in rows and face the camera one row at a time for best results.
> - Good lighting is critical — overhead fluorescent lighting works well.
> - The camera does not need to see the whole class at once; scan row by row.
> - Click "Pause capture" if you need to stop scanning temporarily without closing the session.

6.3 — Manual Attendance Fallback

Sometimes a student's face cannot be recognised — poor lighting, camera angle, or if they were not enrolled properly. In this case:

1. Find the student in the **"Not yet marked"** panel on the right
2. Click **"Mark manually"** next to their name
3. The student moves to the "Marked present" list. Their record is tagged with `method: manual` so it is distinguishable in reports.

6.4 — Closing a Session

Once all students have been captured (or the lecture ends), close the session to lock the register.

1. Click **"Close session"** (red button, top-right of the capture screen)
2. A confirmation dialog appears: *"Close this session? No further attendance can be captured."*
3. Click **OK**. The session status changes to `closed` and you are redirected to the session register.

> 💡 **Closing from the sessions list:**
>
> You can also close any open session from the main Sessions list by clicking the "Close" button in the actions column.

6.5 — Viewing & Correcting the Register

Click any session in the Sessions list (or "View register" link) to see the full attendance register for that session.

- **Present list:** student name, matric number, time marked, method (face/manual), match distance score

- **Absent list:** students not marked
- **Export CSV:** download the register as a CSV file
- **Delete a record:** Admin/lecturer can remove an incorrect attendance record by clicking the delete icon next to it
- **Toggle open/closed:** Admin can reopen a closed session to add missed records, then close it again

7 Attendance Reports

Click **Attendance report** in the sidebar. This page computes the full attendance register for any course, showing every student against every session held.

Selecting a Course

Use the **Course** dropdown to select the course you want to view. The report loads automatically. The URL updates with a `?course=...` parameter so you can bookmark it.

Summary Statistics

Four cards appear at the top of the report:

CARD	WHAT IT SHOWS
Sessions held	Total number of closed sessions for this course
Registered students	Number of students on the course's class list
Average attendance	Mean attendance percentage across all students
Below requirement	Count of students below the required attendance threshold (default 75%). Amber if any at risk, green if none.

The Attendance Grid

The grid table shows each student as a row and each session date as a column:

P — Present **L** — Late **A** — Absent

The final two columns show:

- **%** — Attendance percentage (attended / sessions held × 100)
- **Eligibility** — **Eligible** if percentage ≥ required attendance, otherwise **Below requirement**

Students are sorted from highest attendance percentage to lowest.

Exporting to CSV

1. Click **"Export CSV"** (top-right of the report page)
2. A file named `[COURSE-CODE]-attendance-report.csv` is downloaded
3. Open in Microsoft Excel, Google Sheets, or any spreadsheet app

The CSV includes: S/N, Matric number, Full name, Level, one column per session date (P/L/A), Attended count, Sessions held, Percentage, Eligible (Yes/No).

💡 **Use case — Exam eligibility:**

Run this report at the end of semester. Filter or sort the "Eligible" column in your spreadsheet to identify students who cannot sit for the exam due to low attendance.

8 Staff Accounts (Admin only)

Only Administrators can access this section. The "Staff accounts" link appears in the sidebar only when logged in as an admin.

This page allows you to create and view all staff (lecturer and admin) accounts that can log into the system.

Viewing Existing Staff

The right panel lists all staff accounts with their name, email, department, and role badge (`admin` or `lecturer`).

Creating a New Staff Account

1. Click
   Staff accounts
   in the sidebar

2. **Fill in the "Add staff account" form (left panel):**

FIELD	REQUIRED?	NOTES
Full name	Yes	e.g. `Dr. Amaka Obi`
Email	Yes	Used as the login username. Must be unique.
Password	Yes	Minimum 6 characters. Share securely with the staff member.
Role	Yes	Select **Lecturer** or **Administrator**
Staff ID	No	e.g. `STF/023`
Department	No	e.g. `Computer Science`

3. **Click "Create account".**
   The new account appears in the list on the right.

> ⚠️ **No password reset feature:**
> There is currently no "forgot password" flow. If a staff member forgets their password, an admin must create a new account for them or the database password must be reset manually.

Role Differences — Quick Reference

PERMISSION	ADMIN	LECTURER
Register students / enrol faces	✅ Yes	❌ No
Create courses	✅ Yes	❌ No

PERMISSION	ADMIN	LECTURER
Start lecture sessions	✅ Yes	✅ Yes (own courses)
Capture & mark attendance	✅ Yes	✅ Yes
View attendance reports	✅ Yes (all courses)	✅ Yes (own courses)
Export CSV reports	✅ Yes	✅ Yes
Delete sessions / students	✅ Yes	❌ No
Create staff accounts	✅ Yes	❌ No
View Staff accounts page	✅ Yes	❌ Hidden

9 User Flows by Role

Use Case 1: First-Time System Setup (Admin)

🛠️ **One-time setup before the semester begins**

1. Log in as Admin (`admin@example.edu` / `admin1234`)
2. Go to
   Staff accounts
   → Create accounts for each lecturer. Share credentials securely.
3. Go to
   Courses
   → Create each course (code, title, level, assign lecturer, set required attendance %)
4. Go to
   Students
   → Register each student: fill details + capture 3 face samples per student
5. Go to each
   Course detail page
   → Add the relevant students to each course's class list
6. System is ready for the semester

Use Case 2: Taking Attendance During a Lecture (Lecturer)

📹 **Every lecture workflow**

1. Log in as Lecturer (`lecturer@example.edu` / `lecturer1234`)
2. Go to
   Lecture sessions
   → Select your course, enter venue, set late threshold → Click "Start session & open camera"
3. On the capture screen, click
   "Start capture"
   and allow camera access
4. Point the camera at students — scan row by row. Green boxes = marked, red = unknown/absent
5. For students the camera cannot recognise → click
   "Mark manually"
   in the absent list

6. When attendance is complete → click **"Close session"** to lock the register

Use Case 3: Checking Exam Eligibility (Admin or Lecturer)

📊 End-of-semester eligibility check

1. Go to **Attendance report**
2. Select the course from the dropdown
3. Review the "Below requirement" counter — e.g. "2 students need 75%"
4. Scan the table for students with the **Below requirement** badge
5. Click **"Export CSV"** to download the register for official records or submission

Use Case 4: Adding a New Student Mid-Semester (Admin)

+ Adding a late-registered student

1. Go to **Students** → click "+ Register student"
2. Fill in student details and capture 3 face samples
3. Go to **Courses** → open the relevant course → click "Add students" → search for the new student and add them
4. The student is now included in future sessions and the attendance report (showing absent for all past sessions)

Use Case 5: Correcting a Wrong Attendance Record (Admin or Lecturer)

✏️ Fixing an incorrect record

1. Go to **Lecture sessions** → find the session in the list → click "View register"
2. Find the incorrect record in the Present list
3. Click the **delete (×) icon** next to the record — this removes it

4. If the session is closed and a student was missed: click the "Reopen" toggle to make it open, mark manually, then close again

Use Case 6: Updating a Student's Face Templates (Admin)

🔄 **Student's appearance has changed significantly**

1. Go to **Students** → click the student's name
2. On the profile page, click **"Re-enrol face"**
3. Capture 3 new samples in good lighting
4. Save — old templates are replaced. The student will now be recognised by the new samples going forward.

10 Tips, Troubleshooting & Limitations

Common Issues & Fixes

PROBLEM	LIKELY CAUSE	SOLUTION
"No face detected" during enrolment	Poor lighting or face too far from camera	Move closer (50–80 cm), improve lighting, remove glasses/hats
Camera won't turn on in browser	Browser permissions or insecure context	Click the camera icon in the browser address bar and "Allow". The site must be accessed over HTTPS (already the case on vercel.app).
Student always shows red box (not recognised)	Face not enrolled, or enrollment samples were poor	Re-enrol the student in good lighting (Section 4.4)
Wrong student being recognised	Face threshold too loose, or poor enrollment samples	Re-enrol both students in better conditions. Delete the wrong record from the register.
"This course already has an open session"	You or another lecturer started a session earlier	The system redirects you to the existing open session automatically
Can't start a session — "Register students first"	Course has no students in its class list	Go to the course detail page and add students (Section 5.2)
Login fails with "Invalid email or password"	Wrong credentials or email typo	Check the credentials table in Section 2. Email is case-insensitive.
Session closed before all students captured	Accidental close or forgot to scan back rows	Admin/lecturer can reopen the session from the register page, add missing students manually, then close again
Report shows no data / "No sessions held yet"	No sessions have been closed for that course yet	Run a session and close it. Only closed sessions appear in the report.

Recognition Tips

- **Lighting is the #1 factor.** Well-lit rooms with overhead lights produce the best results. Avoid backlighting (windows behind students).
- **Distance matters.** Faces need to be clearly visible. One camera at the front works well if students come forward row by row.
- **Enrol in the same environment.** If possible, enrol students in the same room / lighting conditions where attendance will be taken.
- **3 samples per student.** The system requires exactly 3 samples at different angles. More angles = more tolerance for head pose variation.
- **Scan frame size.** Don't try to capture the whole lecture hall at once. Move the camera closer, or have students move to a single point for capture.

Known Limitations

LIMITATION	DETAILS
No liveness detection	A printed photo or phone screen showing a student's face could theoretically pass. Not an issue for supervised settings.
Dark or high-contrast environments	The face detector may miss faces in very dim or strongly backlit conditions. Use the manual fallback.
Large classes at distance	A single webcam at the front of a large hall won't reach back rows clearly. Best for small-to-medium classrooms or close-range capture.
No password reset	If a staff member forgets their password, an admin must create a new account for them.
No student portal	Students cannot log in to check their own attendance. This information must be communicated by the lecturer or admin.

Quick Reference — Key URLs

PAGE	URL PATH
Login	`/login`
Dashboard	`/dashboard`
Students list	`/students`
Register student	`/students/new`
Courses list	`/courses`
Lecture sessions	`/sessions`
Attendance reports	`/reports`
Staff accounts	`/staff`

✅ **You're ready!**

This guide covers every feature of the FaceTrack platform. For the demo site, start by logging in as Admin, exploring the pre-seeded students and courses, then start a lecture session and test the live capture screen.